ACM/IDS 104 APPLIED LINEAR ALGEBRA
PRACTICE PROBLEMS FOR LECTURE 14

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Each lecture is accompanied by two practice problems: a somewhat easier, more practical *Problem A*, and a more difficult, more conceptual *Problem B*. The main goal of the practice problems is threefold: to help you better understand the material covered in the corresponding lecture, to help you prepare to solve problems in problem sets and exams, and to accommodate the diversity of students' math backgrounds by providing both easier and more challenging problems. These problems are for self-practice: they will not be graded, and the solutions—posted on Piazza—also illustrate the expected level of rigor for problem sets and exams.

Problem 14A. Principal Component Analysis: Toy Example
As a friendly reminder, whenever we learn (or develop) a new method or algorithm, it is always very useful to explore how the method works in simple, particular cases where the solution to the problem the algorithm is intended to solve is known. This illuminates the algorithm, makes it more intuitively clear, and serves as a common-sense check.

Consider the data consisting of six points in the plane shown in the left panel of Fig. 1. This data set is inspired by the constellation Lyra shown in the right panel of the figure. Compose the data matrix, find the principal components of the data (by hand), and sketch them on the scatterplot.

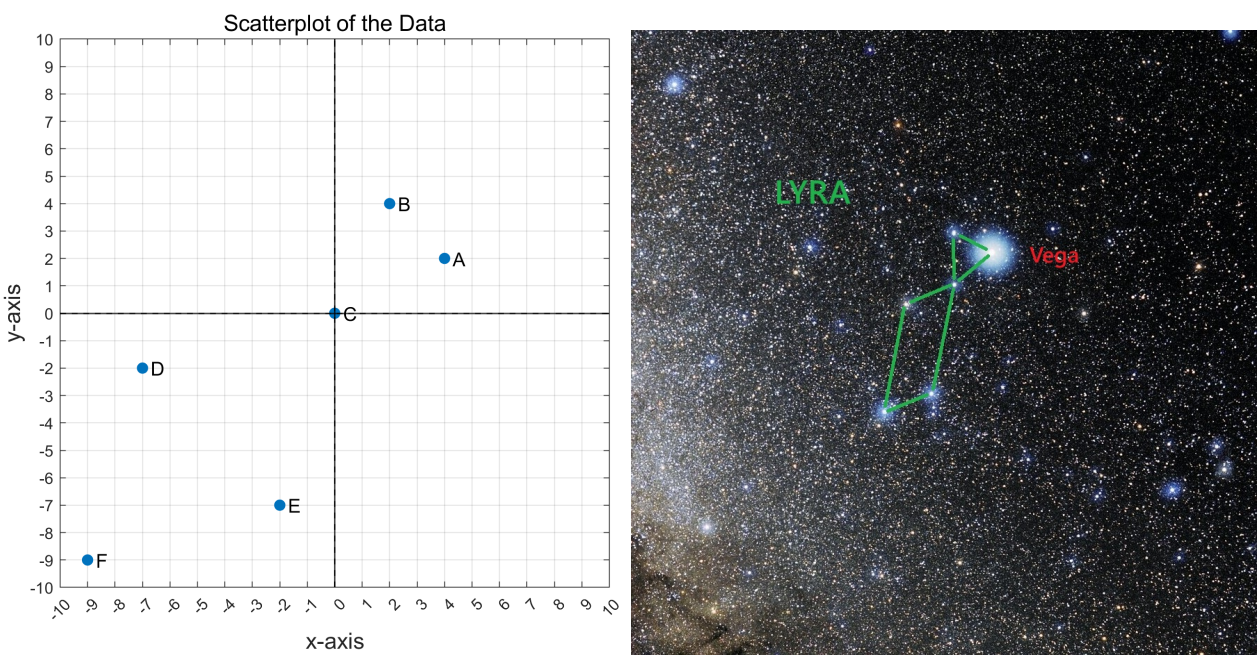

Figure 1. Left panel: Scatterplot of the data. Right panel: Constellation Lyra (Photo by Eckhard Slawik, `https://noirlab.edu/public/education/constellations/lyra/`; minor edits by KZ.). Lyra is a small constellation located in the Northern Hemisphere. The brightest star in Lyra is Vega, which is the fifth-brightest star in the sky. Vega was the pole star in the year 12,000 BC, and, because of the precession of Earth's rotational axis, will again become the pole star around 14,000 AD. In Greek mythology, Lyra represents the lyre of Orpheus. In Wales, Lyra is known as King David's harp.

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Problem 14B. Total Variance of the Data Matrix

Let X be an $m \times n$ data matrix, where columns $x_1, \ldots, x_n \in \mathbb{R}^m$ represent measurements (or observations) and rows $x^1, \ldots, x^m \in \mathbb{R}^n$ represent features (or variables). Suppose that X is centered, that is, each row has zero mean. Geometrically, this means that the cloud of points $x_1, \ldots, x_n \in \mathbb{R}^m$ is centered at the origin. Then, the variance of the ith feature is[1]

$$\mathbb{V}(x^i) = \frac{1}{n} \sum_{j=1}^{n} x_{ij}^2, \quad i = 1, \ldots, m. \tag{1}$$

The *total variance* $\mathbb{V}(X)$ of the data matrix X is defined as the sum of variances of all features.

(a) Let A be an $m \times n$ matrix and B be an $n \times m$ matrix. Show that

$$\operatorname{tr}(AB) = \operatorname{tr}(BA). \tag{2}$$

This means that, although matrices generally do not commute, $AB \neq BA$, they always commute under the trace. This property makes the trace very interesting algebraically, and it is used a lot in a wide range of computations, proofs, and applications.

(b) Prove that the total variance equals the trace of the covariance matrix C_X of the data matrix,

$$\mathbb{V}(X) = \operatorname{tr}(C_X), \qquad \text{where } C_X = \frac{1}{n} X X^T. \tag{3}$$

(c) Let $\lambda_1, \ldots, \lambda_m$ be the eigenvalues of the covariance matrix C_X. Prove that

$$\mathbb{V}(X) = \sum_{i=1}^{m} \lambda_i. \tag{4}$$

[1] If X is not centered, then the variance is $\mathbb{V}(x^i) = \frac{1}{n} \sum_{j=1}^{n} x_{ij}^2 - \left(\frac{1}{n} \sum_{j=1}^{n} x_{ij} \right)^2$, which is less convenient to work with.